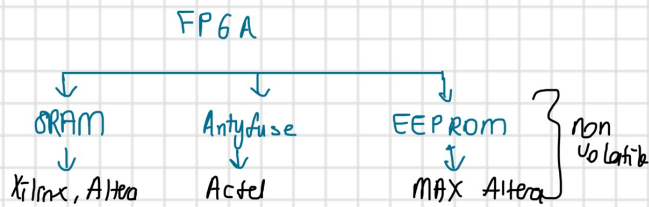


FPGA's

field programmable gate arrays

gate array $\rightarrow$ FPGA / LUT based



SRAM

volatile

External memory

reprogrammability

Quality of systems

technology

power

$\rightarrow$ lifetime is same other digital logical source

$\rightarrow$ programming does not influence

$\rightarrow$ 20 nm dimension of transistor of FPGAs extremely small.

$\rightarrow$ minimum power consumption. The best in this aspect

flasher memory is external chip.

ionization $\rightarrow$ cosmic radiations, -----

high altitude 10 or 11 km
operate aircraft

? filmbed microprocessor technology - semiconductor

too much programmable gate

Maximum power militari is difficult

at home it is high quality for downbands too much videos.

Actel programming element "antifuse" $\rightarrow$ as exactly same with SRAM

LOW power

dimension of active chips is small.

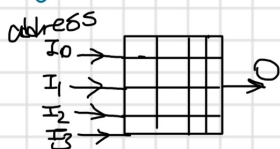
Disadvantages

programming time is very long

Special programmable device

Actel mux instead of LUT

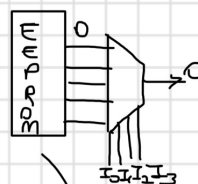
They use mux instead of LUT



We can address some of these input.
with memory

So LUT can memory

Logical functionality is same.



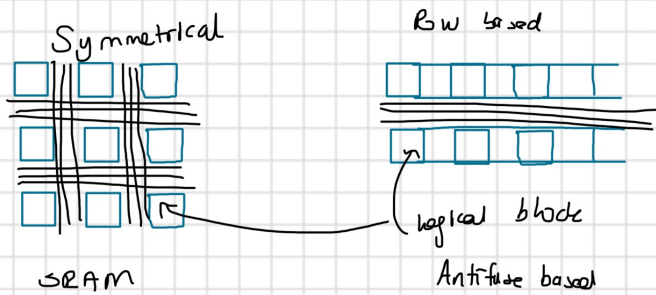
The difference
mux is a
combinational element
that don't have
memory. It can't
remember.

We need an
additional memory
element or we need
to give signals name.

More efficient solutions when use by FPGA
high functionality at low cost

Functions of FPGA Components:

logic blocks
Switching matrices.



Program element are high access in high current row based more selectable

Configuration

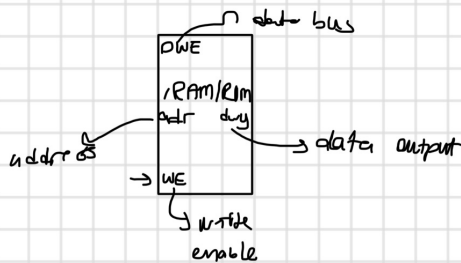
depends on design to antifuse but it too much

EAB ? classic FPGA structure

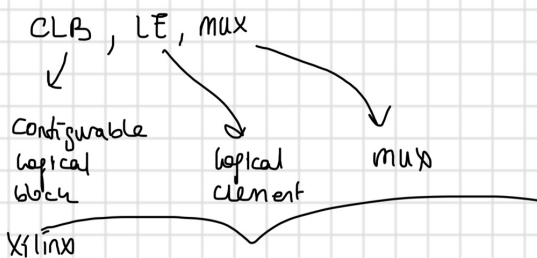
PLL $\Rightarrow$ fast lock loop

Just loop which can generate clock signal (present - other signal could be generate)

Classic embedded memory block in FPGA



local bus $\Rightarrow$
vertical bus

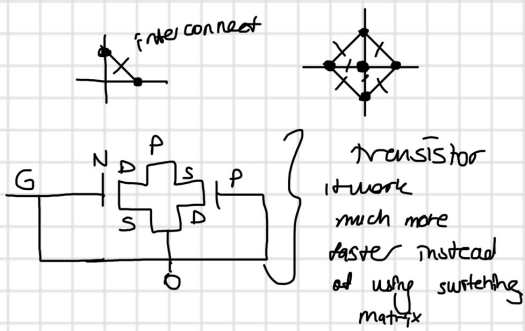


they are same but it changes
from company to company

we have always at least 1 LUT.
the inputs can be changes. Sometimes it
can be more outputs.

Switch matrix

FP6A 900 there are interconnection



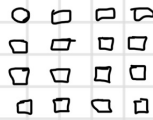
CPLD or FPGA
complex programmable
logical devices

matrix of logical gate
instead of not matrix of
more gates

CPLD



FPGA



heterogeneous structure

array structure

average no of resources

a large amount of resources

imposed connection structure

high flexibility

high speed

average speed

Xilinx

Altera

Stratix ICLB

Power-PC cores

ARM structure

Spartan 6 Resource

In Your design you can observe in Xilinx do

Soft microprocessors implemented in FPGA

MicroBlaze → Xilinx IP most known in Xilinx
↳ very configurable very wide range

with some instruction you can do C language

Atollic free programming environment

we will discuss

Topic of last project other idea for Spartan 6 → by design
ESP32 microcontroller
Cortex
RISC-V based VHDL
buna bak bu dizaynlar
önemli mikrokontrolör aısından